Revenue Architecture: Unit Economics, Financial Model, and Pricing Test

Week 5 | BUS 395: Venture Creation with AI

Student	John Doe (Instructor Example)
Date	June 22, 2026
Venture	Student Move-Out Coordination Platform
Revenue Model	Consignment marketplace with move-in sale

1. REVENUE MODEL SELECTION

Models Eliminated

- **Subscription.** The service runs once a year for two weeks. Nobody subscribes to something they use in May and forget until next May.
- **Freemium.** The service is coordination. Either you match someone or you do not. There is no obvious premium version of a match.
- **Flat buyer-side fee ($30 "Move-In Package").** AI initially recommended this as the primary model. The logic: incoming students spend $250-450 at Walmart, so $30 for pre-claimed furniture is a 90% discount. The problem: it frames departing students as giving items away for free while the platform charges buyers. That asymmetry feels wrong to buyers when they find out the seller got nothing. It also caps revenue at a flat fee regardless of item value.
- **B2B landlord fee.** Landlords spend $800-1,200/year on cleanup, but that cost is baked into existing cleaning contracts. The cleaning crew comes whether there are five abandoned desks or zero. There is no separable line item to pitch against. Zero landlord interviews were conducted. Priya's apartment manager said no immediately on liability grounds.

Model Selected: Consignment Marketplace with Move-In Sale

Departing students leave items in May. The platform holds inventory over the summer. Arriving students buy at a discount, either pre-purchasing over the summer or buying at a move-in sale during the first two weeks of classes. Departing students get paid only if their items sell. The platform takes a 50% cut of each sale.

How money moves:

1. Departing student drops off items in May. Sets a price (or accepts a suggested price based on condition and category). No upfront payment to the seller.
2. Items are photographed, listed, and held over summer in building storage or a partner location.
3. Incoming students browse listings and pre-purchase over the summer, or buy at a move-in sale event in August.
4. When an item sells, the platform takes a 50% cut. The seller receives the other half via Venmo/Zelle.
5. Items that do not sell after 2 weeks are donated. The seller gets nothing, but they were going to throw these items away anyway.

Evidence for:

- Priya spent $450 at Walmart. A used desk at $30-40 (vs. $80 new) is a clear value proposition for the buyer. The buyer's reference price is retail, not $0.
- Departing students currently get $0 for items they throw away. Even $15 back for a desk they were going to trash is found money. Kenji: "I would pay $20-30 to know my stuff was going to someone who needed it." Flip that: he would be even happier getting $15 for it.
- The consignment model eliminates upfront risk. No cash out until items sell. Unsold items are donated, not stored indefinitely.
- A move-in sale event creates urgency and a physical moment where transactions happen. Students are already shopping during that window.
- This is how thrift stores, consignment shops, and Facebook Marketplace already work. Students understand the model.

Evidence against:

- Zero incoming students interviewed. Willingness to buy used furniture at specific price points is untested.
- Holding inventory for 3 months creates real costs: storage space, organization, potential damage. Unlike pure coordination, this model has physical infrastructure requirements.
- A 50% cut may feel high to sellers once they see the math. A desk that sells for $35 means $17.50 to the platform and $17.50 to the seller. But justified by 3-month storage, photography, listing, and running the sale event. And the seller's alternative is $0.
- Three interviewees will not download an app. Payment and listing must run through channels they already use.

> ***Why this beats the flat fee model:*** *The flat fee ($30 per claim) capped revenue and created an asymmetry where sellers gave items for free while buyers paid. The consignment model aligns incentives: sellers get cash for items they were going to trash, buyers get discounts vs. retail, and the platform earns a cut on every transaction. Revenue scales with volume and item value instead of being capped at a flat rate.*

2. UNIT ECONOMICS

Full unit economics with all assumptions are in the attached Excel (Unit Economics sheet). Below is the analysis.

Revenue

The unit of revenue is one item sold. A departing student might drop off 3-5 items. An arriving student might buy 2-4. The average item sells for $30 (ASSUMPTION), based on used pricing at 30-50% of retail. Most items are small or commodity: a lamp at $12, a kitchen starter kit at $20. Desks and bookshelves pull the average up, but the weighted mix lands closer to $30 than $45.

The platform takes a 50% cut (ASSUMPTION). That is at the high end of the 30-50% consignment range, but it covers three months of storage, photography, listing, and running the sale event. The seller gets $15 per item sold. Their alternative is $0 because they were going to throw it away. Even $15 for a desk headed for the dumpster is found money.

Costs

The total variable cost per item is $13 at a $15/hr founder rate. That includes payment processing ($0.50), storage supplies ($2), transport and damage allowance ($3), and 30 minutes of founder time ($7.50). Every number is an assumption. None have been validated with real operations yet.

The critical insight is batching. The flat fee model required 2 hours per customer because every transaction was individual: outreach, matching, handoff. The consignment model batches operations. You collect 20 items in one afternoon, photograph them in one sitting, and sell 30+ at a single move-in sale event. Per-item time drops from 2 hours to 30 minutes. That is the difference between a variable cost of $47.50 (flat fee) and $13 (consignment), a 73% reduction.

Margins

At $15/hr founder time, the margin is $2 per item (13%). Thin but positive. As a student project with labor at $0/hr, the margin jumps to $9.50 per item (63%). The flat fee model lost $17.50 on every customer. The consignment model makes $2. That is a $19.50 improvement per unit, entirely from batching and switching to a percentage cut.

Unsold Items

Not every item collected will sell. Estimated sell-through is 65% in the first season (ASSUMPTION, based on thrift store benchmarks). The remaining 35% get donated after the two-week move-in window. Unsold items still cost $3 each to handle and dispose. In the base scenario, that is 21 unsold items at $63 total, a real drag on the season. Curating what items to accept at intake matters more than collecting volume. Rejecting damaged or low-value items at the door improves sell-through and cuts waste costs.

Cost Audit

The AI's first pass showed 98% margins with $0 for storage, transport, and damage. After three rounds of pushback, realistic costs were added and the margin collapsed to -58% on the flat fee model. The pivot to consignment then restructured the economics entirely: batching cut labor by 75%, the percentage cut replaced the flat fee, and unsold inventory was accounted for. The full audit trail is in the Excel (Cost Audit sheet). The lesson: AI will zero out any cost that seems uncertain. Always run the audit.

3. FINANCIAL MODEL: FIRST SEASON (MAY-AUGUST 2026)

The full financial model is in the attached Excel file (Week5-Financial-Model.xlsx) with unit economics, three scenarios, sensitivity analysis, and cost audit trail. Below is the summary.

Scenario	Items Collected	Sell-Through	Storage	Net (at $15/hr)	Net (at $0/hr)
Pessimistic	30	50%	Rented ($125/mo)	**-$620**	-$400
Base	60	65%	Free (classroom)	**-$120**	+$172
Optimistic	100	75%	Free (classroom)	**-$100**	+$463

The scenarios differ by three assumptions: storage cost (rented vs. free), sell-through rate (50% to 75%), and adoption (8 to 25 departing students). All three lose money at $15/hr founder time. The base case is positive as a student project (+$172).

> **All three scenarios still lose money at $15/hr founder time.** But the losses are dramatically smaller than the flat fee model (-$120 base vs. -$725 base). The path to profitability: reduce per-item time from 30 min to 15 min (margin jumps to $5.75/item), or collect 120+ items at 75% sell-through. Both achievable in season 2.

What This Tells Me

1. **The consignment model is viable as a student project.** At $0/hr founder time, the base scenario generates $172 in cash. The optimistic case generates $463. That is real money and proof the concept works.
2. **Free storage is the make-or-break variable.** Rented storage at $125/month adds $500 and turns every scenario deeply negative. The university classroom is not a nice-to-have. It is the business.
3. **The move-in sale event is the revenue engine.** Pre-purchases trickle in over summer, but the sale event in August is where most transactions happen. This concentrates founder time into one intense week instead of spreading it over 3 months.
4. **Sell-through rate matters more than volume.** Collecting 100 items and selling 50% is worse than collecting 60 and selling 75%. Curating what items to accept matters more than collecting volume.
5. **The path to profitability:** Cut per-item time to 15 minutes through templates, helpers, and process refinement. Or collect 120+ items at 75% sell-through. Both achievable in season 2. The Excel sensitivity tab shows the exact breakpoints.

4. PRICING TEST PLAN

Hypothesis

Arriving international students will buy used furniture at 30-50% of retail price from a campus move-in sale, because it costs them $250-450 to buy new, they lose 1-2 days of orientation to furniture shopping, and the items are right on campus. Departing students will drop off items for consignment because they currently get $0 (trash) and any payout is found money.

Test Method: Two-Sided Price Card

Test both sides of the marketplace with concrete prices.

Steps

Monday-Tuesday: Create two price cards:

Seller card (departing students):

- "Drop off your furniture before you leave. If it sells in August, you get paid."
- Example payouts: desk $30, bookshelf $20, lamp $12, kitchen set $25
- "Items that do not sell get donated. You drop off. We handle the rest."

Buyer card (arriving students):

- "Move-in sale: furnished for less. Used furniture from graduating students."
- Example prices: desk $50 (retail $90), bookshelf $35 (retail $70), lamp $20 (retail $40)
- "Pre-order now. Pick up on move-in day. Or shop the sale the first week of classes."

Wednesday-Thursday: Show the cards to 5 people:

- **Group A (sellers):** 2 departing students I already interviewed (Kenji and Ana volunteered). Show the seller card. Ask: "Would you drop off your furniture for this? What would you keep vs. drop off?"
- **Group B (buyers):** 3 incoming students through ISA WhatsApp group. Show the buyer card. Ask: "Would you buy a used desk for $50 instead of a new one for $90? What about sight unseen, from a photo?"

Friday: Write up results. For each person: what they said, their reaction to specific prices, and whether they would commit now.

Success and Failure Criteria

Signal	What It Looks Like
Yes (sellers)	Both say they would drop off items. At least one says the payout does not matter because they were going to trash it anyway. Neither asks for upfront payment.
Yes (buyers)	2+ out of 3 say the prices are fair. At least 1 asks about specific items or wants to pre-order. The discount vs. retail is the driver, not the concept itself.
No (sellers)	Sellers want upfront payment before drop-off. Or they say the payout is too low to bother. "I would rather just leave it in the hallway."
No (buyers)	Buyers say they would rather buy new. Or they do not trust the condition of used items. "I need to see it first" and there is no way to see it.

If the Test Fails

1. **Increase seller payout:** Drop the platform cut from 50% to 35%. Revenue per item drops to $10.50, but if it unlocks supply, the volume compensates.
2. **Bundle pricing:** Sell "starter packages" (desk + chair + lamp + kitchen kit) at $100-120 flat. Simpler for buyers. Higher average transaction.
3. **Negotiate free storage with the university:** Pitch a locked classroom or storage locker that doubles as the showroom for the move-in sale. If storage is free, the entire cost structure improves.

5. UPDATED VENTURE.MD: REVENUE MODEL SECTION

Chosen model: Consignment marketplace with move-in sale. Departing students drop off items in May. The platform holds inventory, photographs and lists items for pre-purchase over summer, and runs a move-in sale during the first two weeks of August. Sellers get paid only when items sell. The platform takes a 50% cut.

Unit economics: At $30 average item price and a 50% cut, revenue per item sold is $15. Variable cost per item is $13 (including founder time at $15/hr for 30 minutes). Margin is $2 per item at realistic labor rates, or $9.50 per item as a student project. Unsold items (35% of inventory) cost $3 each to handle and donate.

Biggest financial risk: Storage costs. If free holding space (locked classroom or university partnership) cannot be secured, rented storage at $125/month adds $500 to a seasonal model that is already tight. The second risk is sell-through rate. Below 60%, the model loses money even with free storage.

What the pricing test should resolve: Two things. (1) Will departing students drop off items for consignment when the payout is not guaranteed? (2) Will arriving students buy used furniture at 30-50% of retail, either sight-unseen from photos or at a move-in sale? If sellers will not supply and buyers will not buy at these prices, the model needs a fundamentally different structure.